

Task-Routed Video-Language Reasoning Across Traffic Domains: Korea Drive at AI City Challenge 2026

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Abstract. Track 3 of the 2026 AI City Challenge evaluates traffic-video reasoning on three independent leaderboards: in-domain Traffic Anomaly Reasoning (TAR), fisheye traffic-violation understanding (FETV), and egocentric pedestrian-intent VQA (PSI-VQA). We describe Korea Drive, a task-routed framework that combines open Qwen video-language backbones with task-specific prompts, strict output contracts, validation, vocabulary normalization, and a training-derived temporal prior. The official TAR run used Qwen2.5-VL-7B with 4-bit inference; the FETV and PSI-VQA runs used Qwen3-VL-8B through the same software interface. Without task fine-tuning, the system ranked 24th on TAR (0.4256), 3rd on FETV (0.4634), and 5th on PSI-VQA (57.04). Component scores reveal a consistent split: descriptions, cue-style answers, and global scene attributes were competitive, whereas actor identity, lane geometry, position, and temporal grounding were substantially weaker. A duration-stratified prior raised official PSI temporal mIoU from 0.0253 to 0.5708. We release code and artifacts while clearly separating official results from post-deadline analysis.

Keywords: Traffic video · Vision-language models · Domain shift

1 Introduction

Traffic-video understanding increasingly requires more than recognizing that an unusual event occurred. A useful system must identify the relevant actors, follow them through time, explain causal relationships, localize important intervals, and produce outputs that can be evaluated automatically. The 2026 AI City Challenge Track 3 formalizes this setting through three independent leaderboards [1]: (i) in-domain Traffic Anomaly Reasoning (TAR) over CCTV clips, (ii) fisheye Traffic Violation Understanding (FETV), and (iii) Pedestrian Situated Intent VQA (PSI-VQA) over egocentric dashcam video. The latter two are optional out-of-domain evaluations designed to test whether a traffic-reasoning system remains useful after major changes in viewpoint, task structure, and output space.

This paper describes the system submitted by Team Korea Drive (Team 277). The contribution is primarily systems-oriented and diagnostic. A shared framework routes each item to a task-specific prompt, frame-sampling policy, output contract, parser, and optional calibrator. The official submissions did not use one identical checkpoint: TAR used Qwen2.5-VL-7B-Instruct [3], while the two OOD runs used Qwen3-VL-8B-Instruct [2]. We therefore do not claim a controlled checkpoint-level domain-generalization experiment. Instead, we study which capabilities remained robust when a common inference design was reused across three benchmarks.

The system achieved 0.4256 on TAR, 0.4634 and rank 3 on FETV, and 57.04 and rank 5 on PSI-VQA. The released component scores provide unusually clear evidence about strengths and weaknesses. On FETV, our description score of 0.4238 was second-highest overall and exceeded the winning team’s 0.4171, while violation type, positions, and lanes were below 0.18. On PSI-VQA, cue-style OpenQA reached 0.6019, but binary and multiple-choice intent judgments remained far behind the leading systems. A temporal prior fitted only on training annotations achieved 0.5708 official mIoU, indicating that benchmark timing regularities were more reliable than the VLM’s own temporal output.

Our contributions are:

1. A task-routed VLM inference framework spanning TAR, FETV, and PSI-VQA on a single 24 GB GPU, with official ranks of 3rd on FETV and 5th on PSI-VQA.
2. Metric-compatible output contracts, parsing, normalization, and validation for heterogeneous exact-match, language, categorical, and temporal metrics.
3. A controlled evaluation of the PSI temporal prior and a field-level FETV diagnosis showing a large gap between language/global-scene outputs and actor-centric geometry.
4. A reproducibility record separating official artifacts, repository-side candidates, and post-deadline experiments, while documenting missing provenance.

2 Related Work

AI City and traffic-video reasoning. The AI City Challenge has expanded from detection and tracking toward language-based traffic understanding and multimodal reasoning [8]. The 2026 TAR release contains 44,040 reasoning annotations over 3,670 videos, while the test protocol spans ten task types and human-verified evaluation items [1]. This setting makes the benchmark relevant to both multi-task adaptation and robustness under domain and viewpoint shifts.

Video-language models. Qwen2.5-VL supports dynamic-resolution visual processing, native video input, and absolute-time encoding [3]. Qwen3-VL extends the family with stronger multimodal reasoning, long-context processing, and explicit text-based time alignment [2]. Low-bit inference, including NF4 quantization popularized by QLoRA [5], makes 7–8B models practical on consumer

GPUs. Our system freezes the backbone and focuses on routing, prompting, structured generation, calibration, and operational reliability.

Fisheye traffic and pedestrian intent. FishEye8K demonstrates the detection difficulty introduced by radial distortion and nonuniform resolution [6]. FETV extends this domain to structured violation understanding. PSI provides socially situated pedestrian behavior data for autonomous driving [4]; PSI-VQA reformulates part of it as binary, open-ended, multiple-choice, and temporal video QA. Both OOD tasks require persistent target identity and geometry in addition to language generation.

3 Task Setting

3.1 Traffic Anomaly Reasoning

The TAR test set contains 960 items over 80 CCTV clips. Its ten task types include binary and multiple-choice QA, open QA, scene description, causal linkage, temporal description and localization, and summarization. Binary and multiple-choice tasks use regex-based exact match; open-ended tasks use BERTScore-F1 [9]. Following the organizer’s July 1 update, temporal localization remains in the submission file but is excluded from the TAR mean [1].

3.2 Fisheye Traffic Violation Understanding

FETV contains 200 short fisheye clips. Each prediction is a JSON object with a free-text description and twelve structured fields: date, time, violation type, violator type, color, initial and final position, initial and final lane, intersection type, weather, and lighting. The server also reports description and per-field categorical scores. The final score is

$$\begin{aligned} S_{\text{FETV}} = & 0.25 \text{CIDEr}_{\text{norm}} \\ & + 0.25 \text{BERTScore} \\ & + 0.5 \text{MacroF1}. \end{aligned} \tag{1}$$

The evaluation details and output schema are defined by the organizers [1].

3.3 Pedestrian Situated Intent VQA

PSI-VQA uses 40 egocentric dashcam clips and four subtasks: binary crossing-intent classification, cue-style open QA, multiple-choice cue identification, and temporal localization of the driver-decision-critical interval. OpenQA cues are semantically matched using sentence embeddings [7]; the final score equally weights the four normalized task scores [1].

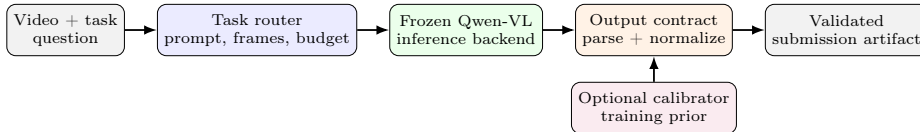


Fig. 1: Task-routed inference framework. The software interface is shared, but the official TAR and OOD runs used different Qwen backbone versions.

Table 1: Official configuration boundary. “Not preserved” denotes metadata that was not logged during the original run.

Evaluation	Backbone and runtime	Benchmark-specific processing
TAR	Qwen2.5-VL-7B-Instruct; NF4 4-bit; task-dependent frame and token budgets	Prompt routing, final-answer extraction, and official format validation.
FETV	Qwen3-VL-8B-Instruct; bf16; at most 16 frames; 151,200 pixels/frame; greedy decoding	Closed vocabularies, JSON parsing, conservative fallbacks, and deterministic description formatting.
PSI-VQA	Qwen3-VL-8B-Instruct through the same interface; original Hub revision not preserved	Cue-format answers and a duration-stratified temporal prior fitted on training annotations.

4 System

4.1 Shared Architecture and Official Configurations

Figure 1 shows the common software architecture. A router selects a prompt program, frame-sampling policy, generation budget, output contract, and parser. Benchmark-specific post-processing then normalizes vocabulary or replaces an unreliable temporal prediction with a training-derived prior. All runners write outputs incrementally, resume from checkpoints, and validate format and coverage before upload.

Table 1 records the official configuration boundary. TAR used Qwen2.5-VL-7B with NF4 4-bit inference and task-dependent budgets. The official FETV run used Qwen3-VL-8B in bf16 with at most 16 frames and a 151,200-pixel budget per frame. PSI-VQA used Qwen3-VL-8B through the same interface; the exact portal-side upload filename and the original Hub revision were not preserved.

4.2 Metric-Compatible Output Contracts

The organizers released validators and metric definitions, so outputs were designed to be unambiguous under those contracts. Binary and multiple-choice items end in a normalized final token; temporal predictions are emitted as bare

JSON; and FETV uses a fixed key set and closed vocabulary. The system extracts the final answer after generation, preventing explanatory text from corrupting exact-match evaluation. For open-ended TAR items, prompts request coverage of setting, actors, event, and aftermath rather than extremely terse answers. For PSI OpenQA, independently verifiable observations are emitted as short bullet points to align with cue-level precision and recall.

This design is not a new reasoning architecture. Its value is operational: heterogeneous tasks can share one inference interface while remaining parseable and reproducible. Every candidate was checked for item coverage, legal labels, key sets, and temporal JSON structure before upload.

4.3 TAR Prompt Routing

The TAR router assigns generation budgets and answer patterns by task. Binary questions use a reason-then-answer instruction: the model briefly tracks the named actors and then emits a final Yes/No token. Multiple-choice tasks similarly end with one letter. Description and summarization tasks use longer budgets to avoid under-specifying events under BERTScore. The backbone was not fine-tuned on the 44,040 released annotations, leaving a clear gap to systems that exploited the training corpus.

4.4 FETV Structured Generation

The official FETV pipeline asks Qwen3-VL for one structured record per clip. Public examples define the 3×3 center-square position convention, direction-relative lane numbering, and report style. A parser retains only the official keys, maps case and vocabulary variants to legal labels, and uses conservative defaults when generation is unparseable. The description opener is formatted deterministically from the predicted date and time, improving consistency without changing the underlying event interpretation.

The official system did not contain the detector–tracker–geometry stack described later in Section 7. We also do not attribute the strong date/time fields to a standalone OCR engine: the released official code used overlay-aware prompting and deterministic parsing/formatting. This distinction aligns the method description with the archived implementation.

4.5 PSI-VQA and the Temporal Prior

The official PSI pipeline uses Qwen3-VL for BCQ, OpenQA, and MCQ generation. OpenQA is normalized to short cue lists. The main additional component is a duration-stratified temporal prior fitted only on the 227 training intervals. Let d denote clip duration. The submitted rule is

$$\hat{I}(d) = \begin{cases} [0.30d, 0.70d], & d < 10 \text{ s}, \\ [0.20d, 0.62d], & d \geq 10 \text{ s}, \end{cases} \quad (2)$$

Table 2: Official full-test results for Team Korea Drive.

Evaluation Rank	Score	Selected component scores
TAR	24	0.4256 BCQ 0.5438; MCQ 0.5875; MCQ-OE 0.7664
FETV	3	0.4634 Description 0.4238; categorical mean 0.5031
PSI-VQA	5	57.0400 BCQ 0.5045; OpenQA 0.6019; MCQ 0.6044; T-mIoU 0.5708

with endpoints rounded to valid integer-second timestamps.

The prior was motivated by an initial official temporal score of 0.0253. After repairing timestamp decoding, the VLM pipeline reached 0.437 mIoU on 60 held-out training items. The pure prior reached 0.571 on the full training set and 0.553 mean mIoU over eight video-grouped 50/50 splits; simple endpoint and center-width blends did not improve it. The final submission therefore used the pure prior and achieved 0.5708 official mIoU. This is a benchmark-specific calibration result, not evidence that temporal priors generally solve pedestrian-intent localization.

4.6 Submission Engineering

Prediction files are written incrementally and can resume after interruption. Missing media receive type-correct fallbacks so coverage remains complete. Candidates are frozen and validated before further experimentation because a schema or upload failure can be operationally indistinguishable from a model failure when the server provides no traceback.

5 Results

5.1 Official Results

Table 2 summarizes the final full-test results. The leaderboards are independent and their scores are not combined. The strongest placements were FETV rank 3 and PSI-VQA rank 5.

5.2 FETV Leaderboard and Field Analysis

Table 3 shows the final FETV table. Our description score was second-highest overall and 0.0067 above the winning team’s description score, but the categorical gap determined the final rank. Pairing our description score with the winner’s categorical mean would yield 0.4925.

The per-field results in Table 4 separate two regimes. Global scene attributes and overlay-derived metadata were strong, while fields requiring selection of one violator and reasoning about that actor’s path were weak. A wrong actor selection can simultaneously corrupt violation type, actor type, color, position, and lane. Fisheye distortion further makes language-only estimates of direction-relative lane geometry unreliable.

Table 3: Final FETV leaderboard (full test set).

Rank	Team	Final Description	Categorical mean	
1	UWIPL_ETRI	0.4891	0.4171	0.5612
2	MR-CAS	0.4884	0.4411	0.5358
3	Korea Drive	0.4634	0.4238	0.5031
4	MobilityAI	0.4525	0.4087	0.4963
5	GDG	0.4209	0.3582	0.4835

Table 4: Official per-field FETV scores.

Field	Score
Weather	1.0000
Lighting	1.0000
Date	1.0000
Time	0.9400
Intersection type	0.7841
Violator type	0.3127
Color	0.2434
Initial / final lane	0.1780 / 0.1694
Violation type	0.1578
Initial / final position	0.1239 / 0.1278

5.3 PSI-VQA Leaderboard

Table 5 shows the final PSI-VQA results. Our OpenQA score exceeded that of the winning team, although it was below the best OpenQA systems. The temporal prior was 0.0043 mIoU below the fourth-place team. BCQ and MCQ were the largest gaps to the top four, consistent with failures in maintaining the identity of the briefly marked target pedestrian.

5.4 Diagnostic Evidence

Table 6 distinguishes controlled training-set diagnostics from submission-to-submission observations. Only the temporal comparison uses the same labeled data under a grouped evaluation protocol. The TAR entries compare intermediate and final leaderboard states and may include concurrent implementation changes; they are therefore not controlled ablations.

6 Cross-Domain Failure Analysis

The results support a consistent, but carefully bounded, interpretation.

Table 5: Final PSI-VQA leaderboard (full test set).

Rank	Team	Final BCQ	mF1	OpenQA	MCQ	Acc.	T-mIoU
1	UWIPL_ETRI	70.64	0.7084	0.5833	0.7912	0.7427	
2	MobilityAI	69.07	0.6136	0.6674	0.7912	0.6906	
3	OptimAI	65.55	0.6464	0.5846	0.7253	0.6656	
4	MR-CAS	64.42	0.5934	0.6389	0.7692	0.5751	
5	Korea Drive	57.04	0.5045	0.6019	0.6044	0.5708	
6	SMART Lab	54.24	0.5796	0.5793	0.6154	0.3955	
7	Team KODE	53.15	0.5796	0.5793	0.5714	0.3955	

Table 6: Diagnostic evidence. The last two rows are submission-to-submission observations rather than controlled ablations.

Comparison	Result
Repaired VLM temporal output vs. duration prior on labeled PSI training data	0.437 vs. 0.571 mIoU
Duration prior over eight video-grouped 50/50 splits	0.553 mean mIoU
Initial official temporal submission vs. final prior submission	0.0253 to 0.5708 mIoU
Short BCQ output vs. reason-then-answer prompting	0.4625 intermediate to 0.5438 final
Terse vs. fuller TAR summary prompting	0.0993 intermediate to 0.3160 final

Relatively robust capabilities. The system produced competitive descriptions, cue-style observations, and global scene attributes. These outputs benefit from pretrained semantic knowledge, explicit contracts, and surface-form normalization. They require less persistent identity maintenance than actor-specific geometric fields.

Weak capabilities. The main failures required binding one entity across time and mapping motion to an external geometric convention. FETV positions and lanes require selecting the violator, following it through fisheye distortion, and interpreting lane indices from the actor’s direction of travel. PSI BCQ and MCQ require maintaining the marked pedestrian’s identity after the annotation disappears. The temporal result adds a separate caution: a strong leaderboard score can arise from annotation regularities rather than video evidence.

Scope of the generalization claim. Because TAR and OOD submissions used different Qwen versions, this work is not a controlled study of one checkpoint’s OOD generalization. In addition, some development decisions were informed by sequential leaderboard feedback. The supported claim is narrower: a shared task-routing and structured-output design reused across domains preserved language-oriented performance better than identity-sensitive spatial and temporal grounding.

7 Post-Deadline Redesign and Analysis

All work in this section was completed after the challenge deadline and has no official score.

For FETV, we implemented a detector-agnostic structured prototype that accepts tracked boxes, lane polygons, event flags, scene metadata, and overlay text. It selects a violator before predicting dependent attributes, computes center-square positions from coordinates, assigns lanes using actor heading, and emits the official schema. Synthetic unit tests cover geometry, lane assignment, violator selection, time parsing, and final record validation. A real-video system still requires an upstream detector, tracker, lane segmenter, and signal-state estimator.

For PSI-VQA, a post-deadline experiment detected the red annotation, re-generated target-aware MCQ answers, and retained the official temporal prior. On a small paired held-out check, MCQ accuracy improved from 3/24 to 9/24, with six corrected items and no regressions in that sample. A two-cue OpenQA prior reached 0.7816 held-out Cue-F1 under pessimistic one-to-one matching. These results are diagnostic only: they were not submitted and must not be interpreted as leaderboard performance.

The redesign suggests a hybrid architecture. Geometry and identity should be handled by detection, tracking, segmentation, pose, and trajectory modules; the VLM should adjudicate ambiguous semantic labels and generate language from a frozen structured record. This division directly targets the field-level failures observed in the official results.

8 Limitations, Reproducibility, and Ethics

Limitations. First, different backbone versions were used across leaderboards, preventing a controlled checkpoint-level transfer claim. Second, some prompt changes were evaluated through sequential submissions rather than isolated experiments. Third, the temporal prior exploits benchmark-specific clipping and annotation statistics and may not transfer to a dataset with different temporal conventions. Fourth, TAR did not use the released 44K annotations for fine-tuning. Fifth, the exact Hugging Face revision, random state, and runtime of the original OOD runs were not preserved. Finally, one TAR source clip was unavailable because of an age-gated upstream video and received a type-correct fallback.

Reproducibility. The project repository contains code and artifact records. It includes prompts, scripts, dataset-layout instructions, environment files, checksums, and a tagged release. The artifact mapping is:

- FETV: `track3_anomaly/submissions/fetv_submission_v11.json`. Its configuration can be reconstructed, but exact byte-level reproduction has not been verified because the original Hub revision and runtime state were not recorded.
- PSI-VQA: `track3_anomaly/submissions/psi_vqa_submission_v7.csv` is the repository-side candidate associated with the final 57.0400 score. The portal upload filename and ID were not retained. The earlier unversioned candidate and the post-deadline v8 file are documented separately.

Ethics. Traffic and pedestrian video can reveal personal behavior and raises privacy, bias, and misuse concerns. We use the data only under the benchmark and source-dataset terms, do not redistribute restricted clips, and do not manually label test data. Pedestrian-intent predictions should be treated as uncertain decision-support signals rather than ground truth about a person’s future action.

9 Conclusion

We presented a task-routed VLM system for the three AI City Challenge 2026 Track 3 leaderboards. Without task fine-tuning, it achieved rank 3 on FETV and rank 5 on PSI-VQA while running on one consumer GPU. The most informative result is not that one VLM solved all tasks, but that performance split sharply by capability: language and global scene attributes were comparatively robust, whereas actor identity, fisheye geometry, lane reasoning, and temporal grounding required explicit structure or benchmark-specific calibration. The released implementation and post-challenge prototype provide a basis for replacing monolithic prompting with a detector–tracker–geometry–language pipeline.

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